

SHREEYA KALEBERE

SOFTWARE ENGINEER • AI/ML ENGINEER • FULL-STACK SYSTEMS

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PROFESSIONAL SUMMARY

High-achieving Computer Science & Engineering undergraduate (**Department Rank 2 / 287, CGPA 9.6 / 10.0**) with practical expertise engineering production-grade full-stack platforms, computer vision pipelines, and intelligent signal/ML systems. Experienced across core computer science foundations (DSA, OOP, OS, DBMS) and modern production technologies (React, Node.js/Express, Python, PyTorch, YOLO, MongoDB, SciPy). Proven track record in national hackathons and academic leadership with a commitment to clean code, data privacy, and robust software architecture.

EDUCATION

D.Y. Patil College of Engineering and Technology

2023 – 2027 (Expected)

Bachelor of Technology (B.Tech) in Computer Science & Engineering

Maharashtra, India

- **Academic Standing:** CGPA 9.6 / 10.0 • Department Rank 2nd among 287 students (Top 0.7%).
- **Relevant Coursework:** Data Structures & Algorithms, Operating Systems, Database Management Systems (DBMS), Object-Oriented Programming (Java/C++), Computer Networks & Security, Machine Learning & AI, Computer Vision, Full-Stack Web Development.

TECHNICAL SKILLS

- **Languages:** Python, Java, C++, JavaScript (ES6+), TypeScript, SQL
- **Full-Stack & Web:** React, Node.js, Express.js, MERN Stack, RESTful APIs, Tailwind CSS, HTML5, CSS3, Socket.io, Three.js
- **AI / ML, Vision & Signals:** PyTorch, YOLO, OpenCV, Signal Processing (SciPy/NumPy, FFT Spectrograms, MFCCs), Scikit-Learn, Spatial Filtering, CLAHE, NLP / Token Parsers
- **Databases & Storage:** MongoDB (Aggregation Pipelines, BSON), PostgreSQL, SQLite, Document Stores, ACID Transactions
- **Tools & DevOps:** FastAPI, Docker, Git, GitHub, Linux/CLI, Vite, Postman, Vercel Serverless Functions

KEY ENGINEERING & RESEARCH PROJECTS

Silent Alarm — Discreet Mental Health Wellness & Silent Alert Platform **Full-Stack / Mental Health & Discreet Assistance**
React • Node.js • Express.js • MongoDB • Conversational NLP • State Management • REST APIs • Modern Web UI

- Architected a discreet, browser-based psychological wellness platform providing contextual conversational assistance, non-intrusive wellness nudges, and secure consent-driven data boundaries for users in sensitive environments.
- Decoupled client interaction layers into modular components (WellnessChatbot, WellnessNudgeToast, WellnessGlossaryModal) with ephemeral session persistence and keystroke cadence/dwell time telemetry.
- Constructed an ephemeral session state engine with gentle toast dispatchers, zero non-essential persistent tracking without user consent, and resilient client-side fallback dialogues delivering sub-second response times.

Road Damage & Pothole Distress Detection

Computer Vision / Deep Learning Research

Python • YOLO • OpenCV • Pothole-600 Dataset • PyTorch • Spatial Filters • CLAHE

- Investigated real-time single-stage object detection architectures (YOLO) paired with OpenCV image enhancement on the Pothole-600 dataset to classify asphalt cavities and alligator cracking with 3-tier distress severity mapping.
- Evaluated Adaptive Histogram Equalization (CLAHE) and edge contour thresholding in OpenCV to eliminate false positive cavity detections caused by severe sunlight variance, wet road reflections, and tree shadow occlusions.
- Integrated GPS coordinate and timestamp logging schemas for municipal maintenance routing; associated technical research paper in progress.

Resume Maker & Enhancer — Intelligent ATS Document Optimization Engine

Full-Stack / Document Engine / ATS NLP

React • Node.js • TypeScript • NLP • Regex Parsers • PDF Generation • ATS Scoring Algorithms • REST APIs

- Engineered a real-time reactive resume builder and enhancer that scores ATS compliance, suggests quantitative impact phrasing, and renders standardized vector PDFs.
- Combined a live dual-pane React editor with server-side document normalization, implementing token parsers that cross-reference industry job descriptions against resume sections to evaluate keyword density.
- Created a unified JSON schema for resume Abstract Syntax Trees (AST) to cleanly separate styling themes from raw career data, ensuring pixel-perfect preview-to-PDF rendering parity across multi-column layouts.

Sonar Fingerprint — Acoustic Signature Classification System Signal Processing / Machine Learning / Acoustic Classification

Python • Signal Processing • NumPy • SciPy • PyTorch • FFT Spectrograms • Scikit-Learn

- Developed a signal processing and machine learning pipeline that applies Fast Fourier Transforms (FFT) and Mel-frequency cepstral coefficients (MFCC) to isolate acoustic echo fingerprints from underwater sonar data.
- Transformed raw time-domain sonar waveforms into 2D time-frequency spectrogram matrices, feeding spectral feature tensors into dense PyTorch neural classifiers.
- Implemented spectral subtraction and bandpass Butterworth filtering in SciPy prior to extracting frequency centroid and energy distribution vectors to eliminate ambient noise and thermal gradient distortions.

Food Safe — Full-Stack Food Safety & Quality Platform

Full-Stack / MERN / Food Safety

React • Node.js • Express.js • MongoDB • MERN Stack • OCR Concepts • REST APIs

- Engineered a centralized MERN web application integrating optical character recognition (OCR) and text parsing concepts to catalog food records, parse ingredient text, and flag potential safety hazards.
- Constructed regex normalizers and token pipelines to sanitize raw OCR text before matching against food additive databases, decoding complex E-numbers.
- Decoupled frontend component state from backend parsing microservices, delivering sub-second response times using optimized MongoDB query indexing.

HONORS, AWARDS & LEADERSHIP

- **Department Academic Rank 2 / 287 (Top 0.7%):** Awarded for continuous academic distinction across CSE curricula (**CGPA 9.6 / 10.0**).
- **Eureka Hackathon — 3rd Prize Winner:** Ideated, designed, and defended a functional software prototype under competitive time constraints.
- **National Talent Search Examination (NTSE):** Qualified **Stage 1 & Stage 2** of India's premier national merit scholarship.
- **Smart India Hackathon (SIH) Competitor:** Selected competitor engineering prototypes solving government ministry problem statements.
- **SOF International Mathematics Olympiad (IMO):** Awarded Merit Distinction for quantitative reasoning and analytical problem solving.
- **ISRO Certification:** Completed satellite remote sensing program on *Climate Change and Aerosol Management* with ISRO.
- **Elected Class Representative:** Primary academic liaison for a cohort of 60+ Computer Science undergraduates.

TECHNICAL IMMERSIONS & FIELD EXPEDITIONS

- **IIT Bombay Techfest (2024):** Participated in robotics exhibits, autonomous drone telemetry, and real-time edge AI demonstrations.
- **Dassault Systèmes Women's Fest (2024):** Selected participant engaging with engineering leads on 3D digital twins and PLM software.
- **Baramati Industrial & Agri-Tech Immersion (2024):** Field study of automated manufacturing plants, sensor networks, and PLC telemetry.